FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XVI

JEE (Main)-2021

TEST DATE: 08-08-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

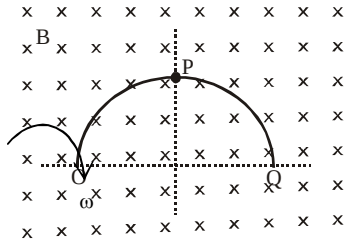
PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A boat having length of 3 metres and breadth 2 metres is floating on a lake. The boat sinks by one cm when a man gets on it. The mass of the man is
   (A) 60 kg
   (B) 62 kg
   (C) 72 kg
   (D) 128 kg

2. The figure shows a semi-circular conducting ring of radius a, which is rotated with constant angular velocity ω about an axis passing through O and parallel to field B as shown. Identify the correct value of potential difference



   (A) $V_P - V_Q = \frac{1}{2}B\omega a^2$
   (B) $V_Q - V_O = 2B\omega a^2$
   (C) $V_P - V_O = \frac{1}{4}B\omega a^2$
   (D) The maximum value of $\left|V_P - V_O\right| = \frac{\pi a^2 \omega B}{2}$

3. Two electrons are moving with the same speed V. One electron enters a region of uniform electric field while the other enters a region of uniform magnetic field. After sometime if the de Broglie wavelength of the two are λ_1 and λ_2, then (select the best alternative)
   (A) $\lambda_1 = \lambda_2$
   (B) $\lambda_1 > \lambda_2$
   (C) $\lambda_1 > \lambda_2$ or $\lambda_1 < \lambda_2$
   (D) $\lambda_1 > \lambda_2$ or $\lambda_1 < \lambda_2$ or $\lambda_1 = \lambda_2$

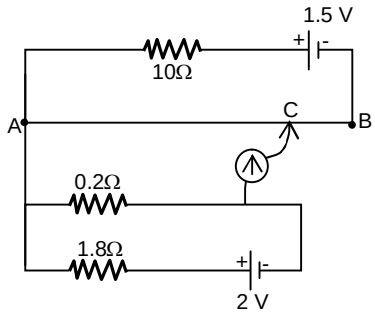
4. The efficiency of Carnot engine is 0.6. It rejects 20 J of heat to the sink. The work done by the engine is
   (A) 20 J
   (B) 30 J
   (C) 33.5 J
   (D) 50 J

5. A standing wave is created in a string which is represented by $y = 6 \sin(100\pi t) \cos(5\pi x)$. If mass per unit length of string is 100 g/m. Then tension in the thread is
(A) 20 N
(B) 100 N
(C) 10 N
(D) 40 N

6. A uniform solid brass sphere is rotating with angular speed ω_0 about a diameter. If its temperature is now increased by 100^0C. What will be its new angular speed (Given: $\alpha_{brass} = 2.0 \times 10^{-5} \text{per}^0\text{C}$)
(A) $1.1\,\omega_0$
(B) $1.01\,\omega_0$
(C) $0.996\,\omega_0$
(D) $0.824\,\omega_0$

7. A train of mass M is moving on a circular track of radius R with constant speed v. The length of train is half the perimeter of track. The linear momentum of the train will be
(A) 0
(B) $2Mv/\pi$
(C) MvR
(D) Mv

8. In the figure, AB is a uniform wire of length 100 cm whose resistance is 5 Ω. If galvanometer reads zero, then AC =
(A) 10cm
(B) 20cm
(C) 40cm
(D) 80cm



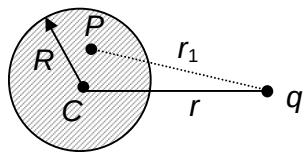
9. There is a 90^0 bend in a pipe line in which water is flowing at speed v. The cross sectional area of pipe is A. The force exerted by pipe walls on water at bend is
(A) $\sqrt{3}\,A\,v^2\rho$
(B) $\sqrt{2}\,Av^2\rho$
(C) $Av^2\rho$
(D) $\sqrt{5}\,Av\rho$

10. A diode as a rectifier converts
(A) alternating current into direct current.
(B) alternating current into pulsating direct current.
(C) high voltage into low voltage.
(D) low voltage into high voltage.

11. A tank has a hole made at its bottom. The time needed to empty the tank from level h_1 to h_2 will be proportional to
(A) $h_1 + h_2$
(B) $\sqrt{h_1} + \sqrt{h_2}$
(C) $h_1 - h_2$
(D) $\sqrt{h_1} - \sqrt{h_2}$

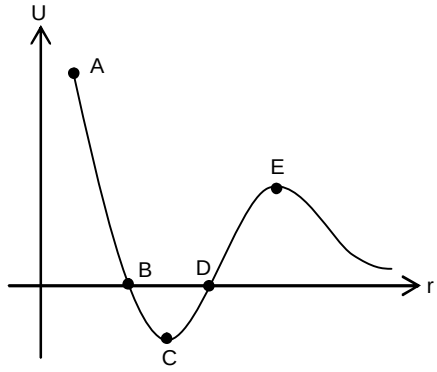
12. At a place the angle of dip is 30°. If the horizontal component of the earth's magnetic field is H, then the total field intensity will be given by
(A) H/2
(B) $2H/\sqrt{3}$
(C) $H\sqrt{2}$
(D) $H\sqrt{3}$

13. A point charge is placed at a distance r from center of a conducting neutral sphere of radius R (r > R). The potential at any point P inside the sphere at a distance r_1 from point charge due to induced charge of the sphere is given by [$k = \frac{1}{4\pi\varepsilon_0}$]



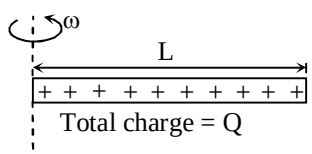
(A) kq/r_1
(B) kq/r
(C) $kq/r - kq/r_1$
(D) $-kq/R$

14. The given plot shows the variation of U, the potential energy of interaction between the particles with the distance separating them, r



(1) B and D are equilibrium points
(2) C is a point of stable equilibrium
(3) The force of interaction between the two particles is attractive between points C and B and repulsive between points D are E on the curve
(4) The force of interaction between the particles is repulsive between C and A
Which of the above statements are correct
(A) 1 and 3
(B) 1 and 4
(C) 2 and 4
(D) 2 and 3

15. A rod has a total charge Q uniformly distributed along its length L. If the rod rotates with angular velocity ω about its end, its magnetic moment is



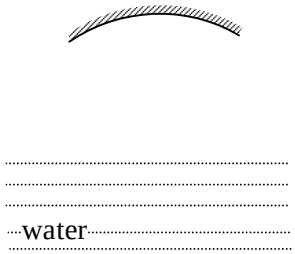
(A) $\frac{Q\omega L^2}{2}$

(B) $\frac{Q\omega L^2}{4}$

(C) $\frac{Q\omega L^2}{8}$

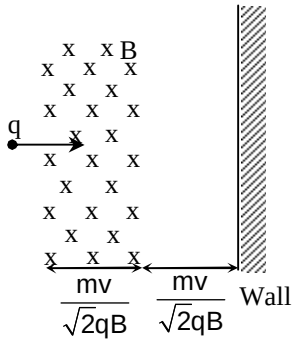
(D) $\frac{Q\omega L^2}{6}$

16. A concave mirror of focal length 15 cm is placed in air at a height of 45 cm above the water surface as shown. The axis of the mirror is vertical and the reflecting surface faces the water surface. A point object is placed on the axis of concave mirror at a distance 35 cm above the water surface. Find the position of image formed after two successive processes- first reflection from the concave mirror and then first refraction from the water (reflective Index of water 4/3). The distance of image from water surface (in m) is



(A) 1
(B) 2
(C) 3
(D) 4

17. A particle of mass m and charge q enters a region of magnetic field (as shown) with speed v. There is a region in which the magnetic field is absent, as shown. The particle after entering the region collides elastically with a rigid wall. Time after which the velocity of particle becomes antiparallel to its initial velocity is



(A) $\frac{m}{2qB}(\pi+4)$

(B) $\frac{m}{qB}(\pi+2)$

(C) $\frac{m}{4qB}(\pi+2)$

(D) $\frac{m}{4qB}(2\pi+3)$

18. If a person walking at the rate of 3km/hour, the rain appears him to fall vertically. When he increase his speed to 6 km/hr it appears to meet him at angle of 30^0 with vertical. The speed of rain is

(A) $3\sqrt{2}$ km/hr

(B) $\frac{3}{\sqrt{2}}$ km/hr

(C) $6\sqrt{2}$ km/hr

(D) 6 km/hr

19. A 3.6 m long vertical pipe is filled completely with a liquid. A small hole is drilled at the base of the pipe due to which liquids starts leaking out. This pipe resonates with a tuning fork. The first two resonances occur when height of water column is 3.22 m and 2.34 m respectively. The area of cross-section of pipe is

(A) $25\ \pi\ cm^2$

(B) $100\ \pi\ cm^2$

(C) $200\ \pi\ cm^2$

(D) $400\ \pi\ cm^2$

20. 1 gm of a radioactive substance takes 50 sec to loose 0.01 gm then the half life of the sample will be

(A) $\dfrac{50\ln(2)}{\ln\left(\dfrac{100}{99}\right)}$

(B) $\dfrac{50\ln(2)}{\ln(100)}$

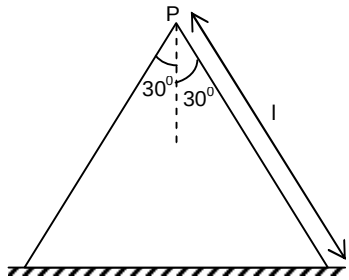
(C) $\dfrac{50\ln(2)}{\ln(99)}$

(D) $\dfrac{50\ln(2)}{\ln(0.99)}$

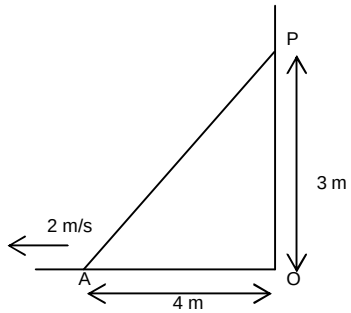
SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. An inverted V is made up of two uniform boards each weighing 200N. Each side has the same length and makes an angle 30^0 with the vertical as shown in figure. The magnitude of the static frictional force that acts on each of the lower end of the V is



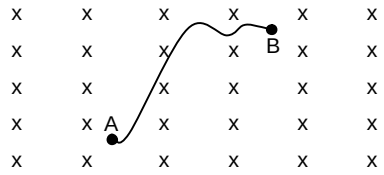
22. A ladder AP of length 5m is inclined to a vertical wall is slipping over a horizontal surface with velocity of 2 m/s when P is at a distance 3m from ground what is the velocity of centre of mass at this moment.



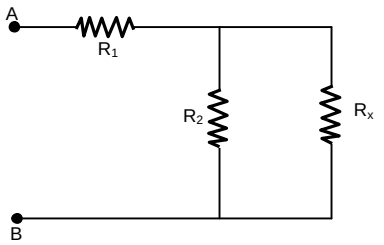
23. A steel rod rigidly clamped at one end free at the other end has longitudinal stationary waves set up in it and it vibrates in a mode in which there are eight displacement nodes other than the rigid end. The young's modulus for steel is 2×10^{11}Pa and its density is 7.8 x 10^3 kg/m^3. The frequency of the source that sets the longitudinal waves in the rod is 10 kHz. The length of the rod is

24. Two electrons are moving in air "side by side" along parallel straight lines at the same speed $V = 3\times10^5 m/sec$. The distance between the electrons is 1 mm. If force of magnetic interaction between the electrons are $m\times10^{-29}N$ Find the value of m.

25. The wire shown in the figure between points A and B carries a current of 10A. A uniform magnetic field of 10T exists in the region which is pointing into the plane in which, wire lies. The coordinates of A and B are (2,0) m and (0,2) m respectively and the length of the wire is 10m. If the magnitude of the force acting on the wire is $m\times100$ N. Find value of m.

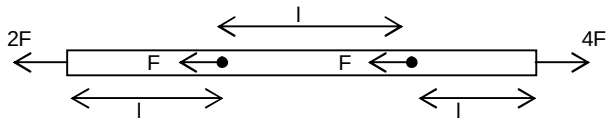


26. A circuit shown in figure has resistance R_1 & R_2. At what value of the resistance Rx will the thermal power generated in it be practically independent of small variations of that resistance? The voltage between the points A and B is supposed to be constant in this case (take $R_1 = 9, R_2 = 3$)



27. A transverse travelling wave represented by equation $y = a\sin(2\pi x - 20\pi t)$, where x is in meters and t is in seconds, is travelling on a string of mass per unit length 0.01 kg/m. Find the kinetic energy of a portion of the string of length equal to one wavelength. ($\pi a = \frac{5}{4}$ unit)

28. Force acting on a uniform rod having length 3l & area of crossection A and young's modulus Y are shown in figure.



Find elongation in the rod (take $\frac{Fl}{YA} = \frac{1}{4}$)

29. A point A is located at a distance r = 1.5 m from a point source of sound of frequency f = 600 Hz. The power of the source P = 0.80 W. Neglecting the damping of the wave and assuming the velocity of the sound in air to be equal to V = 340 m/s. The oscillation amplitude of particles of medium at the point A is $m\times10^{-7}$ & $\rho_{air} = 1.3 kg/m^3$, find m

30. Find the MI of a rod about an axis through its centre of mass and perpendicular to the length whose linear mass density varies as λ = ax where a is a constant and X is the position of an element of the rod relative to its left end. The length of rod is l.(take $al^4 = 99$)

Chemistry

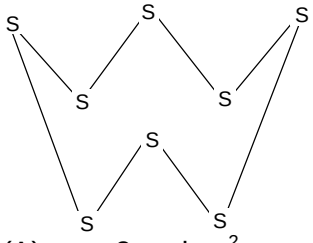
PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which gas shows real behaviour?
(A) 16 g of dioxygen at STP occupies 11.2 L
(B) One gram dihydrogen in 0.5 L flask exerts pressure of 24.63 atm at 300 K
(C) One mole of ammonia at 300 K and 1 atm pressure occupies 22.4 L
(D) 5.6 L of CO_2 at STP is equal to 11 g

32. The oxidation number and hybridization of sulphur in S_8 molecule are respectively:



(A) +2 and sp^2
(B) +2 and sp^3
(C) zero and sp^2
(D) zero and sp^3

33. Which of the following compound on hydrolysis forms products that are polymerized to silicones?
(A) $SiCl_4$
(B) $(CH_3)_2SiCl_2$
(C) SiH_2Cl_2
(D) H_2SiF_6

34. For which of the following aqueous solution, there is maximum difference between the normal and abnormal colligative properties?
(A) 0.1 m $C_6H_{12}O_6$
(B) 0.1 m CH_3COOH
(C) 0.1 m HCOOH
(D) 0.1 m CH_3OH

35. The relation between the molar conductance ($\wedge_m$) and equivalent conductance ($\wedge_e$) of the solution of $MgCl_2.KCl.6H_2O$ is
(A) $\wedge_m = \wedge_e \times 8$
(B) $\wedge_m = \wedge_e \times 3$
(C) $\wedge_e = \wedge_m \times 3$
(D) $\wedge_e = \wedge_m \times 8$

36. $Pb(NO_3)_2\ (excess) + CrO_2Cl_2 + NaOH \xrightarrow{25^oC} Products$

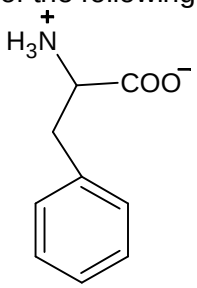
Which of the following precipitates are formed in the above reaction?

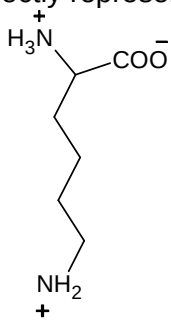
(A) $PbCrO_3$ and $PbCl_2$
(B) $PbCrO_4$ and $PbCl_2$
(C) $PbCr_2O_7$ and $PbCl_4$
(D) Na_2PbO_2 and $PbCl_4$

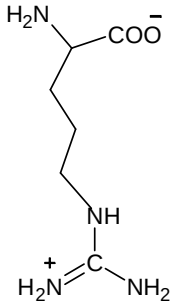
37. How many effective number of Na^+ ions are present in the unit cell of NaCl if the ions along all the body diagonals of the unit cell are removed?

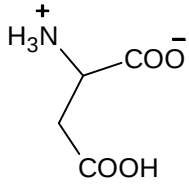
(A) 4
(B) 3
(C) 2
(D) 1

38. Which of the following zwitter ionic form is not correctly represented for an amino acid?

(A) 

(B) 

(C) 

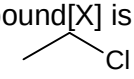
(D) 

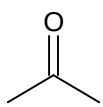
39. Which undergoes dehydration on heating?

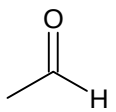
(A) LiOH
(B) NaOH
(C) KOH
(D) RbOH

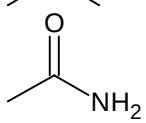
40. $[X] \xrightarrow[(ii)\ H^+(aq)]{(i)\ NH_4Cl;KCN(equimolar)} CH_3CH(NH_2)COOH$

Compound[X] is

(A) 

(B) 

(C) 

(D) 

41. Which of the following two compounds of potassium evolve gases when react with water?
(A) KNO_3 and K_2CO_3
(B) KO_2 and KH
(C) $KHCO_3$ and KOH
(D) K_2O_2 and K_2SO_4

42. $CH_3 - C \equiv C - CH_3 \xrightarrow{Na/\text{liquid } NH_3} (P)$

$(P) \xrightarrow{Br_2/CCl_4} Q$

$(P) \xrightarrow{MnO_4^- / OH^- / \text{Cold}} R + S$

Which of the following statement is correct about the products of above reaction?
(A) 'Q' contains no asymmetric carbon atom
(B) 'P' and 'Q' are functional isomers
(C) 50% of R and 50% of Q makes a racemic mixture
(D) 'P' has one stereoisomer

43. $CH_3CH = CH - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH = CHCH_3$

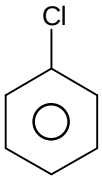
How many stereoisomer(s) is/are possible for the above compound?
(A) 2
(B) 3
(C) 4
(D) 6

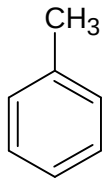
44. $SrCO_3(s) \rightleftharpoons Sr^{2+}(aq) + CO_3^{2-}(aq)$

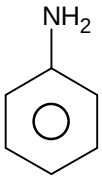
Addition of which of the following substance can increase the solubility of $SrCO_3$ according to above reaction?
(A) NaCl
(B) HCl
(C) K_2CO_3
(D) NaOH

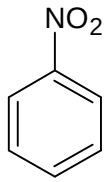
45. Which of the following is not identical between inorganic benzene and organic benzene?
(A) Number of atoms
(B) Number of electrons
(C) Number of protons
(D) Number of polar covalent bonds

46. Which is most reactive towards $CH_2 = CH_2/H^+$ for para substitution?

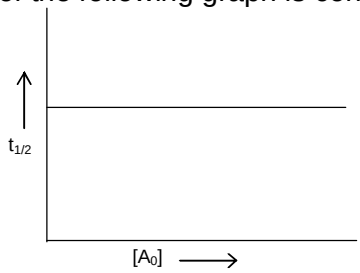
(A) 

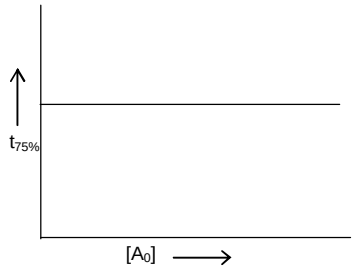
(B) 

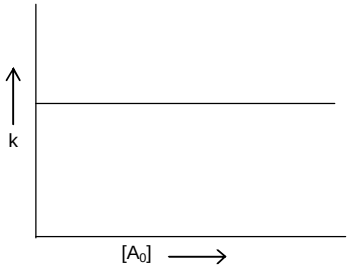
(C) 

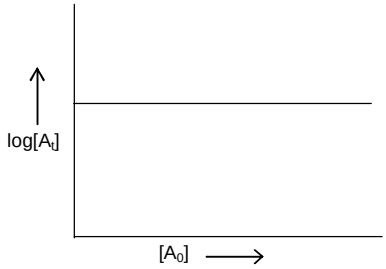
(D) 

47. Which of the following graph is correct for a zero order reaction?

(A) 

(B) 

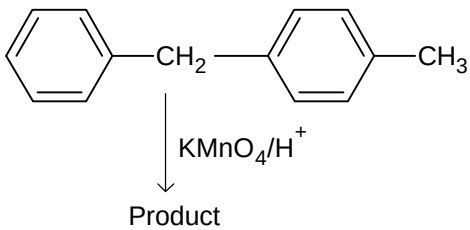
(C) 

(D) 

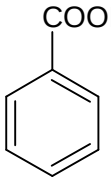
48. What is the pH of 0.1 M NaH_2PO_4?

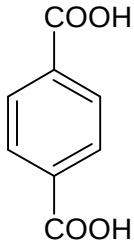
[$p^{K_{a_1}}$, $p^{K_{a_2}}$ and $p^{K_{a_3}}$ of H_3PO_4 are 7.2, 10.6 and 12.8]

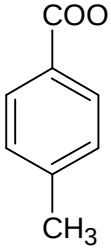
(A) 9.7
(B) 8.01
(C) 10.6
(D) 8.9

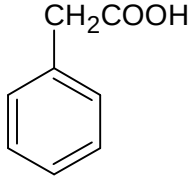
49. 

Which is not a product of above reaction?

(A) 

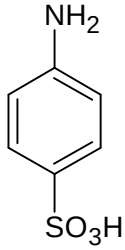
(B) 

(C) 

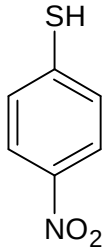
(D) 

50. The sodium extract of which compound does not give red colouration with Fe^{3+} solution?

(A) $HSNH_2NH_2OH$

(B) 

(C) $O_2NCH_2CH_2SH$

(D) 

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. 5 mole of a mixture of $Na_2C_2O_4$ and Na_2CO_3 required 400 mL of 0.5 M acidified $KMnO_4$ for complete oxidation. What is the mole fraction of $Na_2C_2O_4$ in the mixture?

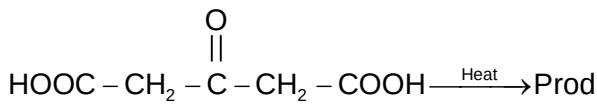
52. What is the root mean square(r.m.s) velocity of CH_4 in ms^{-1} unit if the value of RT is 120 J mol^{-1}?

53. The successive ionization energies of an s-block element are
8.6, 12.9, 1630.2, 2406.2 eV, etc
If atomic mass of the elements is 39.8 g mol^{-1}, what will be the molar mass of it's normal oxide?

54. How many S – O single bonds are present in Caro's acid (H_2SO_5)?

55. The radiator of a car contains 50.5 g water. If 6.2 g of glycol ($HOCH_2CH_2OH$) is added to it, the solution freezes at -4^oC. How many gram of ice will be separated in the process? [K_f = 1.86 K kg mol^{-1}]

56. How many moles of CH_3I can be absorbed by one mole of the following compound?
$H_2N - CH_2 - CH_2 - NO_2$

57.

What is the molar mass of the organic product formed in above reaction in g mol^{-1} unit?

58. $H_3PO_3 \xrightarrow{\text{Heat}} (X) + PH_3$
If oxidation number of phosphorus in compound(X) is +x, what is the value of x?

59. What is the pH of a buffer solution which contains CH_3COOH and CH_3COONa in 1 : 10 molar ratio respectively?
(K_a of $CH_3COOH = 10^{-5}$)

60. How many optical isomer(s) is/are possible for fructose?

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Let $f(xy) = f(x)\, f(y)\ \forall\ x > 0, y > 0$ and

$f(1+x) = 1 + x\{1 + g(x)\}$, where $\lim_{x\to 0} g(x) = 0$, then $\int \frac{f(x)}{f'(x)}dx$ is

(A) $\frac{x^2}{2} + c$

(B) $\frac{x^3}{3} + c$

(C) $\frac{x^2}{3} + c$

(D) $\frac{x^3}{4} + c$

62. Let $x = (\sin\alpha)^{\log_b(\sin\alpha)}$

$y = (\cos\alpha)^{\log_b(\cos\alpha)}$

$z = (\sin\alpha)^{\log_b(\cos\alpha)}$

whose $0 < b < 1$ and $0 < \alpha < \frac{\pi}{4}$, then

(A) $x < z < y$

(B) $y < z < x$

(C) $z < x < y$

(D) $x < y < z$

63. If $f(y) = e^y, g(y) = y,$ and $F(t) = \int_0^t f(t-y)\, g(y)dy,$ then

(A) $F(t) = te^{-1}$

(B) $F(t) = 1 - te^{-1}(1+t)$

(C) $F(t) = e^t - (1+t)$

(D) $F(t) = te^t$

64. Let $f(x) = \frac{1+\sin 2x - \cos 2x}{2}$. If f(p) = f(q)= f(r) = 0 where $0 \le p \le q \le r \le \pi$, then $(p+q+r)$ has the value equal to

(A) $\frac{3\pi}{4}$

(B) $\frac{5\pi}{4}$

(C) $\frac{6\pi}{4}$

(D) $\frac{7\pi}{4}$

65. Let S be the set of points (a,b) in the co-ordinate plane, where each of a and b may be -1,0 or 1. How many distinct line pass through at least two members of S

(A) 8
(B) 20
(C) 24
(D) 27

66. If PQ is a double ordinate of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that OPQ is an equilateral triangle, O being the centre of the hyperbola. Then the eccentricity e of the hyperbola, satisfies

(A) $1 < e < \frac{2}{\sqrt{3}}$

(B) $e = \frac{2}{\sqrt{3}}$

(C) $e = \frac{\sqrt{3}}{2}$

(D) $e > \frac{2}{\sqrt{3}}$

67. $\int \frac{x^9 \ dx}{(4x^2+1)^6}$ is equal to

(A) $\frac{1}{5x}\left(4+\frac{1}{x^2}\right)^{-5} + C$

(B) $\frac{1}{5}\left(4+\frac{1}{x^2}\right)^{-5} + C$

(C) $\frac{1}{10x}\left(1+4x^2\right)^{-5} + C$

(D) $\frac{1}{10}\left(4+\frac{1}{x^2}\right)^{-5} + C$

68. If a, b, c are in G.P., x and y be the A.M.'s between a, b and b, c respectively, then $\left(\frac{a}{x}+\frac{c}{y}\right)\left(\frac{b}{x}+\frac{b}{y}\right)$ is equal to

(A) –2
(B) –4
(C) 2
(D) 4

69. In ΔABC, if $A = 30^o$ and $\frac{b}{c} = \frac{2+\sqrt{3}+\sqrt{2}-1}{2+\sqrt{3}-\sqrt{2}+1}$, then the measure of $\angle C$, is

(A) 75°
(B) 15°
(C) $52\frac{1}{2}^o$
(D) $97\frac{1}{2}^o$

70. A function f has domain $[0,2]$ and range $[0,1]$. Then the domain and range, respectively, of the function defined by $g(x) = 1 - f(x+1)$ are

(A) $[-1,1]$, $[-1,0]$
(B) $[-1,1]$, $[0,1]$
(C) $[0,2]$, $[-1,0]$
(D) $[1,3]$, $[-1,0]$

71. The quadratic equation $ax^2 + bx + c = 0$ where a, b, c are the sides of ΔABC, and the equation $x^2 + \sqrt{2}x + 1 = 0$ have a common root. The measure of $\angle C$, is

(A) 30°
(B) 45°
(C) 60°
(D) 75°

72. 2 circles of radii r_1 and r_2 both touching the coordinate axes, intersecting each other orthogonally. The ratio $\frac{r_1}{r_2}$ equals $(r_1 > r_2)$ is

(A) $2+\sqrt{3}$
(B) $\left(\sqrt{3}+1\right)$
(C) $\sqrt{5}+1$
(D) $2+\sqrt{5}$

73. If the middle points of sides BC, CA and AB of trianlge ABC are respectively D, E, F then position vector of centroid of triangle DEF, when position vector of A, B, C are respectively $i+j$, $j+k$, $k+i$ is

(A) $\frac{1}{3}(i+j+k)$

(B) $(i+j+k)$

(C) $2(i+j+k)$

(D) $\frac{2}{3}(i+j+k)$

74. A function f such that f'(a) = f"(a)=f"'(a)=.....=$f^{(2n)}$(a) = 0 and f has a local maximum value b at x = a, if f(x) is

(A) $(x-a)^{2n-2}$

(B) $b-1-(x+1-a)^{2n+1}$

(C) $b-(x-a)^{2n+2}$

(D) $(x-a)^{2n-2}-b$

75. If $b+c=3a$, then the value of $\cot\frac{B}{2}\cot\frac{C}{2}$ is equal to _____

(A) –18

(B) 18

(C) 2

(D) none of these

76. Total number of common tangents of $x^2+y^2-2x-4y=0$ and $x^2+y^2-8y-4=0$, is equal to :

(A) 1

(B) 2

(C) 4

(D) 3

77. If $\log_{10}\left(ab^3\right)=\log_{10}\left(a^2b\right)=1$, then the value of $\log_{10}(ab)$ is equal to

(A) $\frac{5}{3}$

(B) $\frac{3}{5}$

(C) $\frac{2}{5}$

(D) $\frac{1}{5}$

78. If $\lim\limits_{x\to 0}\dfrac{\sqrt[2010]{1+x}-1}{2x+x^2}=\dfrac{L}{1005}$, then the value of L is

(A) $\dfrac{1}{4.(2009)!}$

(B) $\dfrac{1}{2.(2009)!}$

(C) $\dfrac{1}{8}$

(D) $\dfrac{1}{4}$

79. Three numbers are chosen at random without replacement from 1, 2, 3,…..,10. The probability that the minimum of the chosen numbers is 4 or their maximum is 8, is

(A) $\dfrac{11}{40}$

(B) $\dfrac{3}{10}$

(C) $\dfrac{1}{40}$

(D) none of these

80. If $\begin{vmatrix} 3^2+k & 4^2 & 3^2+3+k \\ 4^2+k & 5^2 & 4^2+4+k \\ 5^2+k & 6^2 & 5^2+5+k \end{vmatrix}=0$, then the value of k is

(A) 2
(B) 1
(C) –1
(D) 0

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If the difference between mean and mode is 63, the difference between mean and median is:

82. Compute the square of the value of the expression $\dfrac{4+\sec 20^o}{\operatorname{cosec} 20^o}$

83. Let $f(x) = x - \frac{1}{x}$. Then find the number of different solutions to the equation $f(f(f(x))) = 1$.

84. If the area of the region bounded by the x – axis and the graph of $y = |x| + |x-1| - k(k > 0)$ is 2 unit, then k^2 is

85. $I_1 = \int_0^1 \frac{e^x}{x+1} dx$, $I_2 = \int_0^1 \frac{(x^2)}{e^{x^3}(2-x^3)} dx$, then $\frac{I_1}{eI_2}$ is equal to

86. If $\frac{z_2}{z_1}$ is pure imaginary, then $\left|\frac{6z_1 + 8z_2}{4z_2 + 3z_1}\right| =$

87. If the total number of selections that can be made from n different books each having 5 copies taking at least 2 copies of each book is 256, then n =

88. The positive integral value of n such that $1.2^1 + 2.2^2 + 3.2^3 + 4.2^4 + ...n.2^n = 2 + 2^{n+5}$, is

89. Find the value of 'a' for which area bounded by $x = 1, x = 2, y = 6x^2$ and $y = f(a)$ is minimum

90. If radius of the incircle of a triangle with its sides 5p, 6p and 5p is 6, then p is equal to: